Replication of Azzouz et al. (2026): a Koch–Sierpinski hybrid fractal antenna

Abdul Jabbar Khan, ECE, RGUKT Nuzvid · Simulation-only replication report, version 2 · 8 Nov 2026

1. Summary

Target paper. Azzouz, Bouhmidi, Munir, Nasralla, Chetioui, *Fractal Fract.* 2026, 10, 47 (doi:10.3390/fractalfract10010047). A seven-band antenna on FR4 (60 × 40 × 0.8 mm) with reported gains up to 19.38 dBi.

Scope of every statement below. The paper does not state the ground plane, port, radiation box, copper thickness or excitation power, and it gives the outline only as a figure. All results are for the setup I assumed (Section 2). “Not reproduced” means my model disagrees under assumptions the paper does not confirm. It does not show that the paper is wrong.

Main findings

- **Resonance frequencies: reproduced.** All seven lie within 1.0 % of the paper’s values (largest difference 0.95 %), each slightly lower. The agreement depends on the port model: a smaller wave port moves the dips down by up to 1.63 % (Section 3.2).
- **S11 depths: partly reproduced.** At 2.11 GHz I obtain −5.8 dB (reported −14 dB). At 8.48 GHz S11 is −12.6 dB (my nearest dip is −16.0 dB at 8.45 GHz; reported −30 dB). My 9.18 and 9.54 GHz dips lie in one −10 dB band.
- **Iterations 0–2 (Fig 4): reproduced** in dip frequency (within 2 %) and approximately in depth. The mismatch is specific to the final geometry.
- **Fig 7 (Koch-only variant): not reproduced.**
- **Gain: not reproduced.** Over 1–10 GHz my peak gain is at most +4.45 dBi, my peak directivity at most 9.28 dBi, and my radiation efficiency 1.5–34.8 %. A dielectric-loss energy balance is consistent with these efficiencies.
- **Fig 9 (patterns):** the paper’s Fig 9 levels agree with its text gains; mine are 6–23 dB lower. Shape comparison is inconclusive.
- **The paper’s gain figures disagree with each other.** Text, Fig 9 and Fig 10 agree; Fig 11 is 1.7–13.7 dB lower at the seven resonances.
- **No tested setup change closes the S11 depth gaps** (eight cases, Section 3.2).

Evidence labels. *Verified* means computed from files or pages. *Inference* and *assumption* are labelled where used. Digitized paper values carry about ±1 dB uncertainty, more at narrow notches and sharp peaks.

2. Scope, method and assumptions

- **Software:** Ansys HFSS 2025 R2.4 Student, one core (the paper used v15). The Student licence caps the mesh at about 40,000 tetrahedra: the largest solved mesh had 40,666 and a 74,904-tetrahedron mesh failed.
- **Geometry:** rebuilt from the paper's Figs 1–2 to about 0.05 mm. The paper gives no base side *a*, no outward shift for the fourth iteration and no carpet ratio. The final outline has 48 vertices (extent ±18.243 × ±19.061 mm). The carpet has 73 slots (one 5 mm, eight 5/3 mm, sixty-four 5/9 mm; outer extent 13.889 mm against D2 = 13.88 mm). Overlay checks of the geometry on Figs 1–2 pass (64 checks).
- **Iteration sequence (from Fig 2):** Iter0 is one downward triangle; Iter1 adds a 5 mm slot; Iter2 is a hexagram with the slot and eight smaller slots; Iter3 is the final Koch-style outline with the 73-slot carpet.
- **Digitization:** pixel extraction with calibrated axes from raster figures (±1 dB). Fig 9 radii are raw pixels, calibrated separately.

Assumed setup (the paper is silent on all of these)

Item	Used
Substrate	FR4, εr = 4.7, tan δ = 0.0197, 0.8 mm, board 60 × 40 mm (from the paper)
Ground plane	Full, 60 × 40 mm, Perfect E
Conductors	Perfect E, zero thickness
Port	Wave port 10 × 6 mm at the feed end, renormalised to 50 Ω; “Use Radiation Boundary on Ports” unchecked
Radiation box	37.5 mm padding, radiation boundary (+X side not padded)
Solver	Multi-frequency adaptive mesh at 2.11, 5.78, 9.56 GHz, up to 15 passes, ΔS 0.02; interpolating sweep 1–12 GHz in 0.01 GHz steps

Solver convergence (all solves met the ΔS 0.02 criterion)

Design	Tetrahedra	Passes	Final ΔS
Iter0	18,408	10	0.016389
Iter1	18,121	11	0.018925
Iter2	21,781	11	0.017419
Iter3 (final)	39,514	11	0.017235
Koch-only	28,736	15 (limit reached)	0.018746

The far-field sweep (1–10 GHz, 0.1 GHz step, 91 points) reuses the final-design adaptive mesh. Far-field values are sampled on a 5° (φ) × 1° (θ) grid.

3. Results by figure

Paper item	Status	Key result
Figs 1–2 geometry	Reconstructed	Fig 2 sequence is the one consistent with Fig 4
Fig 4 iterations	Iter0–2 match; Iter3 depths differ	Dip frequencies within 2 %
Fig 5 S11	Simulated curve compared	Resonances agree, depths do not; measured curve is not a target
Fig 6 mesh	Not replicable	Meaning of 0.005 / 0.0025 / 0.00125 mm not stated
Fig 7 Koch-only	Not reproduced	Claimed wide band and depths not found
Fig 8 VSWR	Compared	2 of 7 values agree within 0.1
Fig 9 patterns	Levels not reproduced; shapes inconclusive	Mine 6–23 dB lower
Fig 10 3D gain	Peak values compared; not reproduced	
Fig 11 gain vs frequency	Not reproduced	Correlation 0.22 with my peak gain
Fig 12 currents	Not compared numerically	Palette mapping medium-low confidence; excitation power unknown
Table 1	Not a replication target	

3.1 S11 at the seven resonances (final design)

Paper f (GHz)	Paper RL (dB)	Paper S11 at f, digitized (dB)	My dip f (GHz)	Difference	My dip depth (dB)	Mine at paper f (dB)	My −10 dB bandwidth (MHz)
2.11	−14	−14.8	2.09	−0.95 %	−5.8	−5.6	0
3.06	−12	−12.3	3.04	−0.65 %	−22.2	−12.0	41

Paper f (GHz)	Paper RL (dB)	Paper S11 at f, digitized (dB)	My dip f (GHz)	Difference	My dip depth (dB)	Mine at paper f (dB)	My −10 dB bandwidth (MHz)
5.78	−16	−16.2	5.76	−0.35 %	−13.3	−11.6	56
6.94	−10	−11.0	6.92	−0.29 %	−13.4	−13.1	183
8.48	−30	−29.9	8.45	−0.35 %	−16.0	−12.6	104
9.23	−16	−16.2	9.18	−0.54 %	−28.6	−16.7	575
9.56	−17	−17.3	9.54	−0.21 %	−37.6	−23.6	575

- The paper's bandwidths are 90, 37, 67, 100, 90, 130 and 220 MHz. Mine differ, and my 9.18 and 9.54 GHz dips share one −10 dB band (9.09–9.66 GHz), which the paper reports as two.
- Narrow notches make depth very sensitive to small frequency offsets. My 3.04, 9.18 and 9.54 GHz dips are deeper than the paper reports; my 2.11 and 8.48 GHz depths are shallower.
- RMSE between my curve and the paper's digitized curve (1.5–10 GHz, clipped at −30 dB) is 2.92 dB.

3.2 Setup sensitivity

Each case changes one item of the baseline (the lumped-port case changes two).

Case	2.1 GHz dip (dB)	8.4 GHz dip (dB)	Largest dip-frequency difference to the paper	Solve (tetrahedra / passes / ΔS)
Baseline (wave port 10 × 6 mm, Fw 1.8 mm, box 37.5 mm)	−5.8	−16.0	0.95 %	39,514 / 11 / 0.017235
Feed width 1.5 mm	−7.6	−16.5	2.74 % (shifts up to 170 MHz)	not recorded
Feed width 1.73 mm (width drawn in the figures)	−6.1	−15.8	⩽ 1 %	not recorded
Radiation box 20 mm	−6.0	−16.0	⩽ 1 %	not recorded

Case	2.1 GHz dip (dB)	8.4 GHz dip (dB)	Largest dip-frequency difference to the paper	Solve (tetrahedra / passes / ΔS)
Finite-conductivity copper (HFSS default, 58,000,000 S/m, zero thickness)	−6.6	−15.0	0.95 %	40,666 / 11 / 0.014311
Wave port 7 × 4 mm	−5.8	−16.1	1.63 %	17,531 / 6 / 0.018536
Lumped port 1.8 × 0.8 mm, +X side padded 37.5 mm	−6.0	−14.7	about 0.5 %	36,938 / 10 / 0.016881

- **Over all cases the 2.1 GHz dip is −7.6 to −5.8 dB (paper −14) and the 8.4 GHz dip −16.5 to −14.7 dB (paper −30).** None of these variables closes the depth gaps.
- **The dip frequencies depend on the port.** The 7 × 4 mm port run also stopped on a coarser mesh, so port size and mesh refinement are not separated.
- **The lumped-port run is confounded:** the port type and the +X padding changed together (the lumped port could not be solved without the padding). Solving it with “Use Radiation Boundary on Ports” checked or unchecked gave identical S11.
- Narrow-notch depths of the lumped run differ from the wave-port baseline (9.54 GHz: −25.3 against −37.6 dB), with frequencies within 10 MHz.

3.3 Fig 4: successive iterations

Iteration	RMSE (dB)	Correlation	Dips ⩽ −10 dB, paper vs mine
0	1.71	0.83	2 vs 2
1	2.84	0.78	3 vs 5
2	2.64	0.83	5 vs 5
3 (final)	2.80	0.82	7 vs 6

RMSE uses the flat-extension method over 1.5–10 GHz clipped at −30 dB; the overlap-only method gives 1.71 / 2.84 / 2.64 / 2.80 dB.

- **Verified:** every paper dip at or below −10 dB for Iter0–2 has one of mine within 2 % (largest 1.7 %). The paper's own Fig 4 curves show its stated resonance counts (2, 3, 5, 7) at the −10 dB level.
- **Inference:** the final-geometry mismatch may come from the unknown outline dimensions (base side, outward shift), to which narrow deep notches are sensitive. This was not tested.
- **Fig 2 against the text:** the text describes iteration 1 as two superimposed triangles, whereas Fig 2(a) shows one triangle. The Fig 2 sequence reproduces Fig 4 in dip frequency, so I used Fig 2 (inference from one simulation per iteration).

3.4 Fig 7: Koch outline without slots

The Fig 7 claims are from the paper's text; the figure was not digitized.

Paper claim	My Koch-only model
Dips near 2.0 and 2.9 GHz, about −10 dB	Nearest dips 2.17 GHz (−6.5 dB) and 3.07 GHz (−23.9 dB)
5.8–6.2 GHz merge into a band of nearly 1000 MHz, better than −25 dB	5.93 GHz dip of −16.1 dB; band 75 MHz wide at −10 dB
7.2 GHz dip below −20 dB	No dip within 3 % of 7.2 GHz at that depth (nearest 6.95 GHz, −12.7 dB)
9.7 GHz, −11 dB, 700 MHz	9.99 GHz dip, 236 MHz wide
(not claimed)	Nothing deeper than −25 dB below 9.5 GHz

The tolerances for this comparison were set with earlier results in view, so they are not blind. The solve reached the 15-pass limit and converged on the last allowed pass (ΔS 0.018746).

Inference (one simulation): in my model much of the multiband behaviour comes from the outline. Iter2, which has the carpet but not the Koch outline, shows a deep 3.8 GHz resonance, so the carpet matters in that geometry.

3.5 Fig 8: VSWR

f (GHz)	2.11	3.06	5.78	6.94	8.48	9.23	9.56
Paper text	1.45	1.65	1.37	1.80	1.06	1.37	1.31

f (GHz)	2.11	3.06	5.78	6.94	8.48	9.23	9.56
Mine	3.20	1.67	1.71	1.57	1.61	1.34	1.14

Only 3.06 and 9.23 GHz agree within 0.1. My 2.11 GHz band does not meet the paper's own VSWR ⩽ 1.8. My VSWR equals the HFSS VSWR column.

3.6 Gain, directivity and efficiency (Figs 9–11)

f (GHz)	Paper text (dBi)	Paper Fig 9 cut max (dBi)	Paper Fig 11 (dBi)	My cut max (dBi)	Text minus mine (dB)	My peak D (dBi)	My efficiency	Aperture yardstick (dBi)
2.11	3.92	3.7	−2.5	−2.32	6.2	4.84	19.3 %	1.7
3.06	8.24	8.6	6.5	−14.81	23.0	2.90	1.5 %	5.0
5.78	6.90	5.6	−6.8	−0.97	7.9	8.82	9.7 %	10.5
6.94	11.66	11.0	2.2	+1.61	10.0	6.62	31.2 %	12.1
8.48	19.38	19.3	13.7	−2.70	22.1	7.81	8.9 %	13.8
9.23	16.76	16.7	10.6	−6.15	22.9	5.88	10.8 %	14.6
9.56	12.06	12.0	5.0	−0.06	12.1	6.51	25.2 %	14.9

- **My cut max is a lower bound.** I exported only four φ planes (0, 90, 180, 270°). At 9.18–9.56 GHz the true peaks are 0.4–2.5 dB higher (HFSS report values: −3.69 dBi at 9.23 GHz and +0.33 dBi at 9.56 GHz).
- **Directivity and efficiency** are taken at the nearest 0.1 GHz sweep point (2.1, 3.1, 5.8, 6.9, 8.5, 9.2, 9.6 GHz). HFSS report tables at the exact frequencies, not stored in the repository, give efficiencies of 19.1, 1.7, 10.3, 32.2, 8.7, 11.1 and 25.1 %.
- **Aperture yardstick:** $4\pi A/\lambda^2$ for the 40 × 60 mm board. It is a plausibility yardstick, not a bound. The paper's 19.38 dBi at 8.48 GHz exceeds it by 5.6 dB, and FR4 loss would reduce gain further.
- **Sweep over 1–10 GHz:** maximum peak gain +4.45 dBi (7.4 GHz), maximum broadside gain +2.67 dBi (7.5 GHz), maximum peak directivity 9.28 dBi (7.5 GHz), efficiency 1.5–34.8 % (mean 14.4 %). Gain equals directivity × efficiency to 0.0002 dB.
- **Peak directions:** near broadside at 2.11 and 5.78 GHz; tilted 33–79° toward φ = 180° at 3.06–9.23 GHz.
- **Energy balance:** the dielectric loss integrated over the substrate matches accepted minus radiated power to 0.1 % at 2.11 GHz and 0.05 % at 3.06 GHz (computed in HFSS, not stored in the repository). My efficiencies are internally consistent at those two

frequencies.

- **Fig 11 is not a fixed-direction gain.** I tested broadside gain against Fig 11 and found correlation −0.03; for peak gain the correlation is 0.22 (mean offset 11.0 dB, RMSE 13.6 dB over 91 points).
- **Within the paper:** the text gain exceeds the digitized Fig 11 value by 1.7–13.7 dB at the seven resonances. The Fig 11 maxima (about 21 dBi near 7.8 GHz and 17 dBi near 4.1 GHz) occur where the paper's S11 is only about −4 dB and −1 dB.
- **Unresolved anomaly:** my peak directivity is below 0 dBi (impossible) at 17 of 91 frequencies (1.0–1.4 and 3.3–4.4 GHz). The cause is unknown. The paper's seven resonances are not among these frequencies, but 3.06 GHz lies just below the affected range, so nearby values carry some doubt.
- **Dependence on assumptions:** gain depends on the assumed ground, port and box. Good S11 agreement for Iter0–2 does not validate gain.

3.7 Fig 9: radiation patterns

- **Angle convention (verified):** the paper's E- and H-plane curves coincide at 90° and 270° in every panel, so broadside lies at the top and bottom of the plots. The paper does not define the angle or its sign.
- **Levels (verified, ±1 dB):** the calibrated peaks of the paper's Fig 9 equal its text gains within 1.3 dB in all seven panels. Mine are 6–23 dB lower (Section 3.6).
- **Shapes (inconclusive):** I compared my four principal cuts with the paper's curves under 16 angle conventions. With a mismatched-frequency null (the paper curve against my cuts at the six other frequencies), the matched frequency is the best of seven for only 2 of 14 curves. Chance expectation is 2.0 ($P(X \geq 2) = 0.61$). The matched value is below the null maximum for 12 of 14 curves and below the null median for 7 of 14. With only the conventions consistent with the z-axis constraint, the matched frequency is never the best.
- **Limit:** only four φ planes were exported, and a 20 dB level gap makes shape agreement unlikely anyway.

3.8 Fig 12: surface currents

Not compared numerically. The palette mapping has medium-low confidence and the paper's excitation power is unknown.

4. Points where the paper is unspecified or its figures disagree

1. **Setup not stated:** ground plane, port, radiation box, mesh, sweep, copper thickness, excitation power and outline dimensions.
2. **Fig 5:** the measured and simulated curves differ. Read by eye, the measured S11 has dips near 2.14, 3.20, 6.03, 7.41, 8.19 and 9.90 GHz and none at 8.48, 9.23 or 9.56 GHz, while the text describes close agreement. No measured gain is shown, although the text mentions agreement for the first four resonances.
3. **Gain:** text, Fig 9 and Fig 10 agree with each other but not with Fig 11 (Section 3.6).
4. **Fig 2 against the text:** iteration 1 is described as two triangles but drawn as one. Fig 2(c) is a hexagram with a two-level carpet.
5. **Fig 6:** the mesh sizes 0.005, 0.0025 and 0.00125 mm giving identical curves cannot be interpreted without a definition of the quantity.
6. **Fig 7 and Table 1:** the Koch-only variant is shown with about 1 GHz bandwidth better than −25 dB, while Table 1 lists 220 MHz as the hybrid's widest bandwidth.
7. **Feed impedance:** the closed-form (Hammerstad) estimate for Fw = 1.8 mm on 0.8 mm FR4 is about 44 Ω, not 50 Ω (my calculation; not confirmed in HFSS). The feed drawn in the paper's figures measures about 1.70 mm; whether this is real or a drawing effect cannot be decided.

5. Checks and open results

- **Overlay checks:** the digitized curves lie on the drawn curves of the original figures (68 checks) and my geometry lies on Figs 1–2 (64 checks).
- **Far-field symmetry:** the φ = 90° and 270° cuts deviate more above 9 GHz, which may indicate mesh noise (inference).
- **Iter0–2 horizontal edges:** in Fig 2 these edges lie about 0.08 mm inside my geometry. The effect on S11 was not tested.
- **Iter1 notch:** my Iter1 shows a narrow (about 20 MHz) −37 dB notch at 8.74 GHz where the paper curve shows none. Whether it is a real resonance or an interpolation artefact is undecided.

6. Limitations

- The full ground, Perfect-E conductors, 50 Ω wave port and 37.5 mm box are assumptions. Gain depends on them.

- The outline dimensions are my reading of Figs 1–2. Narrow deep notches are sensitive to them.
- The Student licence caps the mesh at about 40,000 tetrahedra, which limits refinement. Far-field data above about 9 GHz look noisier.
- Gain cuts cover four φ planes only, so cut maxima are lower bounds.
- This is one solver, one assumed setup and one simulation per design. No independent solver such as openEMS has been run.

7. Open items

1. Discrete solves at 8.70, 8.74 and 8.78 GHz to decide the Iter1 notch.
2. Find the cause of the negative directivity at 17 of 91 frequencies.
3. Separate port type from +X padding with a wave-port run on a padded box.
4. Cross-check the final geometry in openEMS.

8. Reproducibility

Every number above has a source file and script in the companion repository (`koch-sierpinski-antenna-replication`). The claims table (`docs/claims_vs_evidence.md`) lists each statement with its evidence file, script, status and confidence. `python scripts/run_all.py` regenerates all tables and figures.